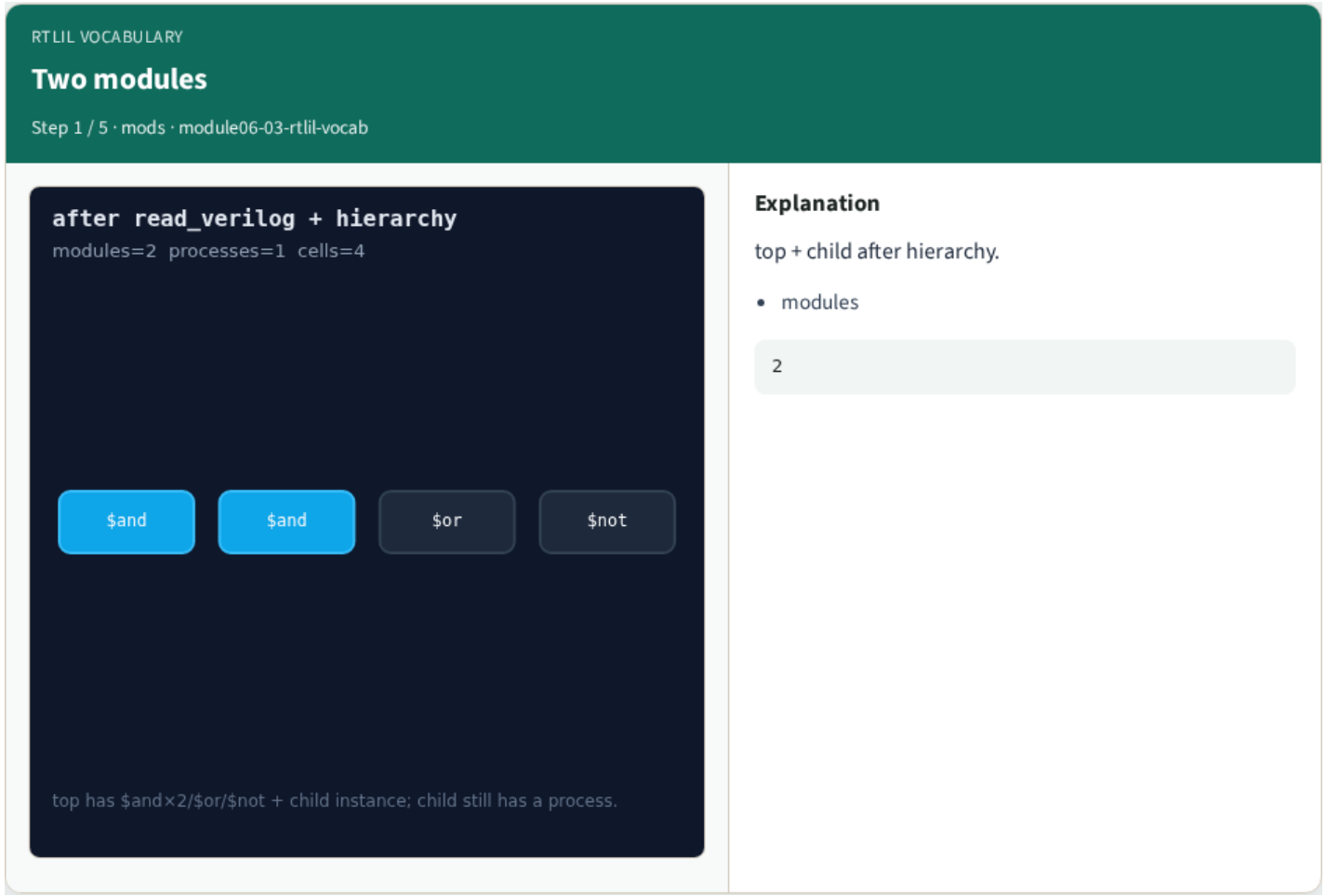
Why rtlil

Yosys stores designs in RTLIL , modules, wires, processes, and $-typed cells you see in dump/stat

The idea

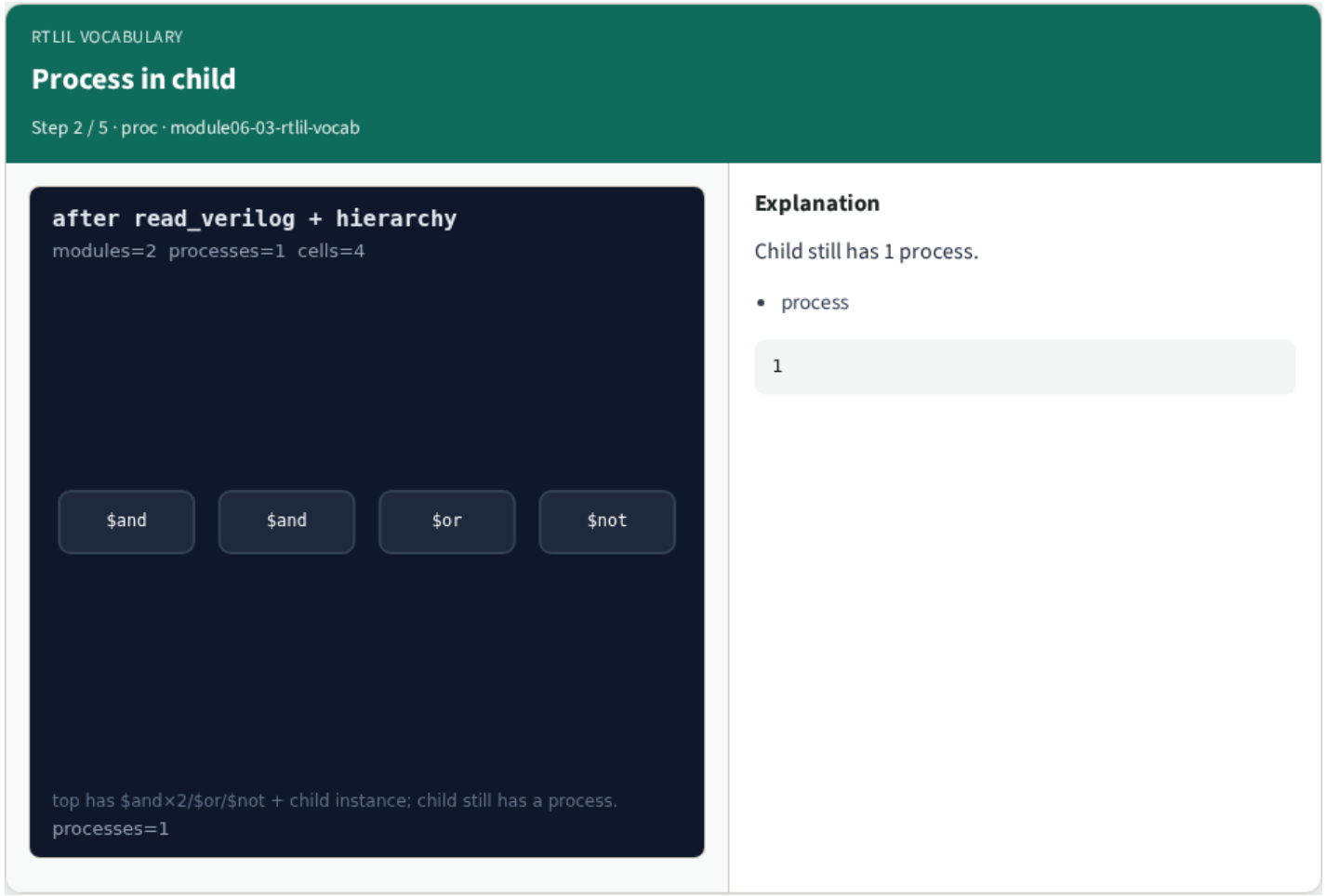
- GOLDEN: after_read modules=2 processes=1 and_count=2; vocab includes $and $or $not $dff

Two modules



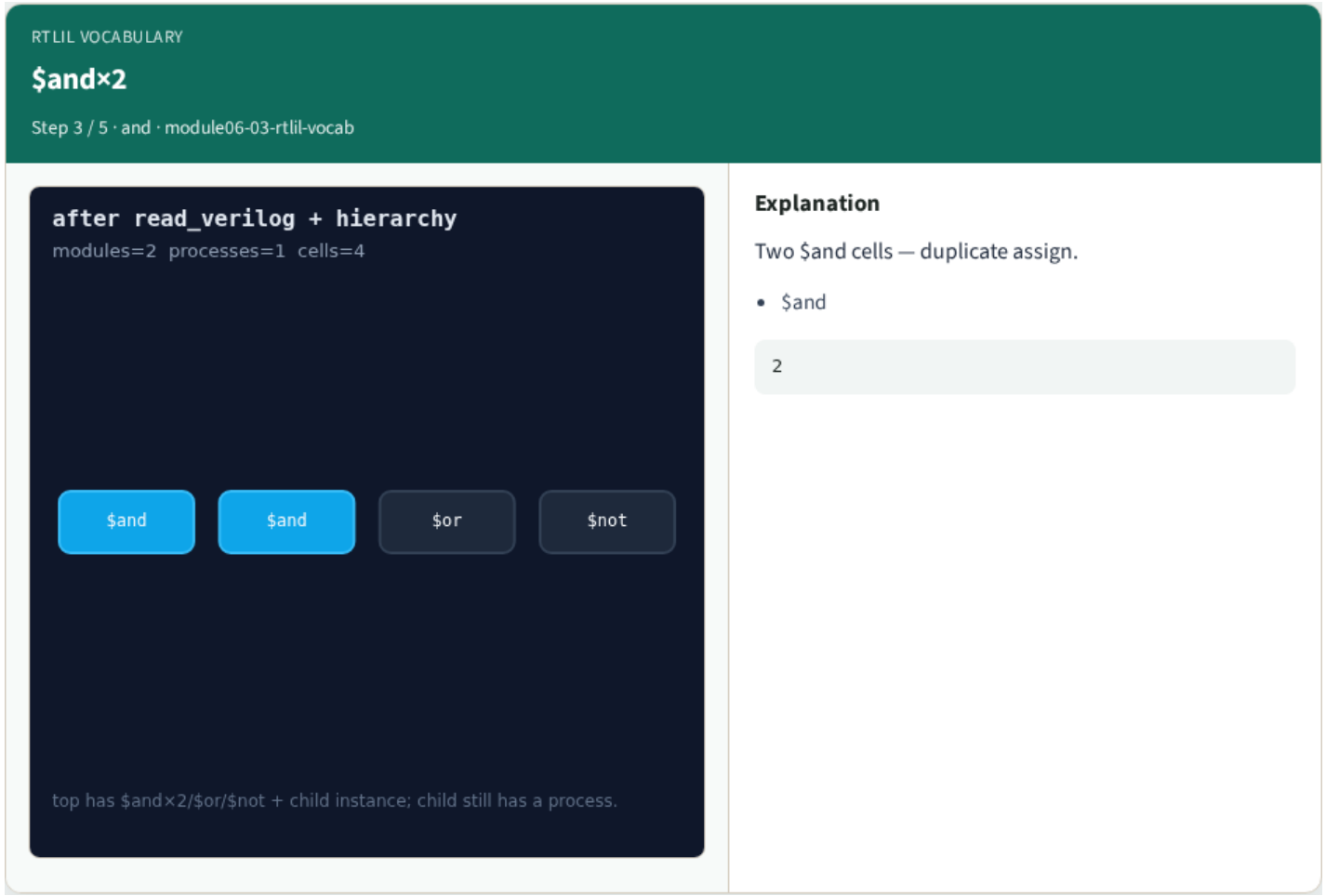
Two modules

Process in child



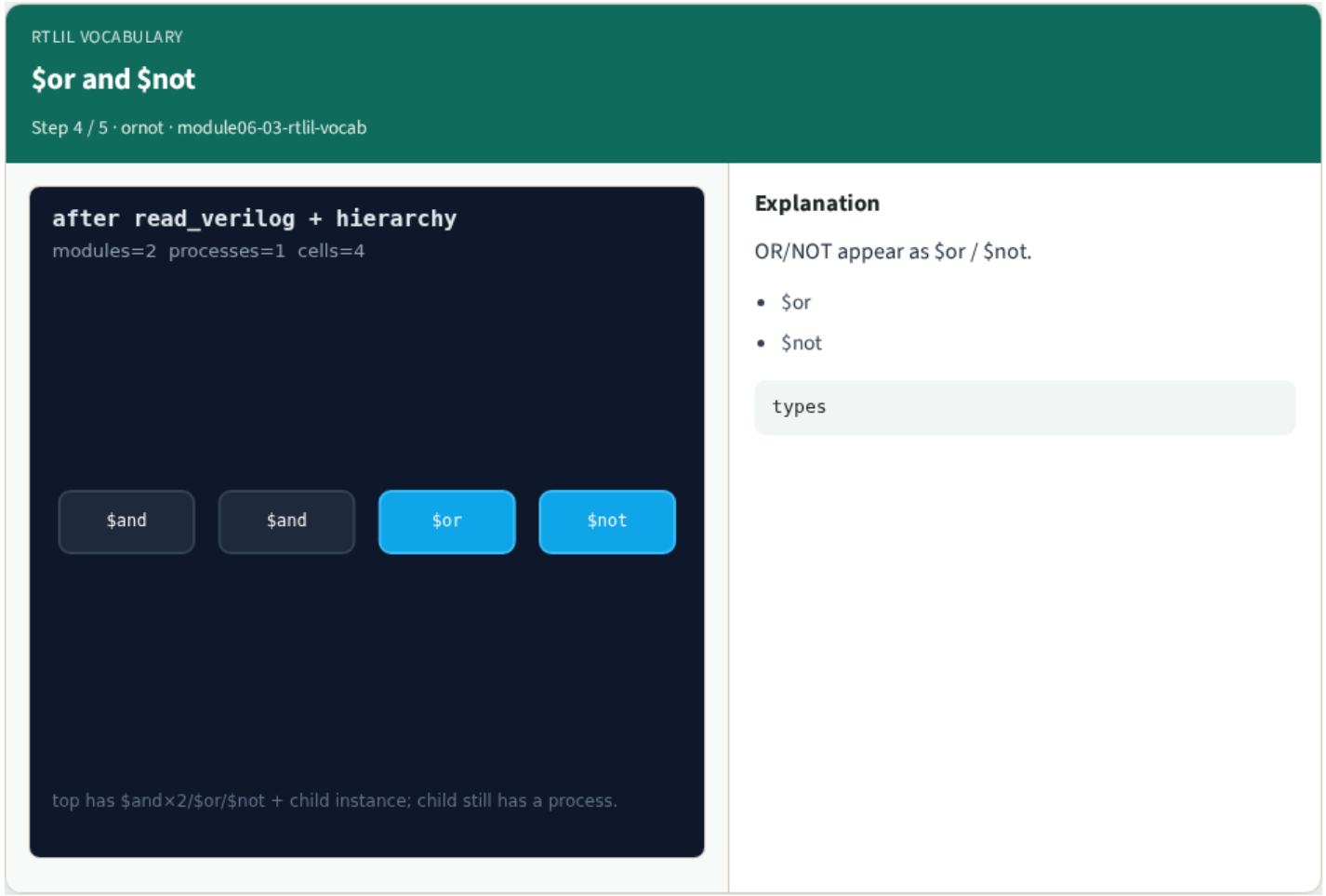
Process in child

$and×2


RTLIL VOCABULARY
$and×2
Step 3 / 5 · and · module06-03-rtlil-vocab
after read_verilog + hierarchy
modules=2 processes=1 cells=4
$and
$and
$or
$not
top has $and×2/$or/$not + child instance; child still has a process.
Explanation
Two $and cells — duplicate assign.
$and
2

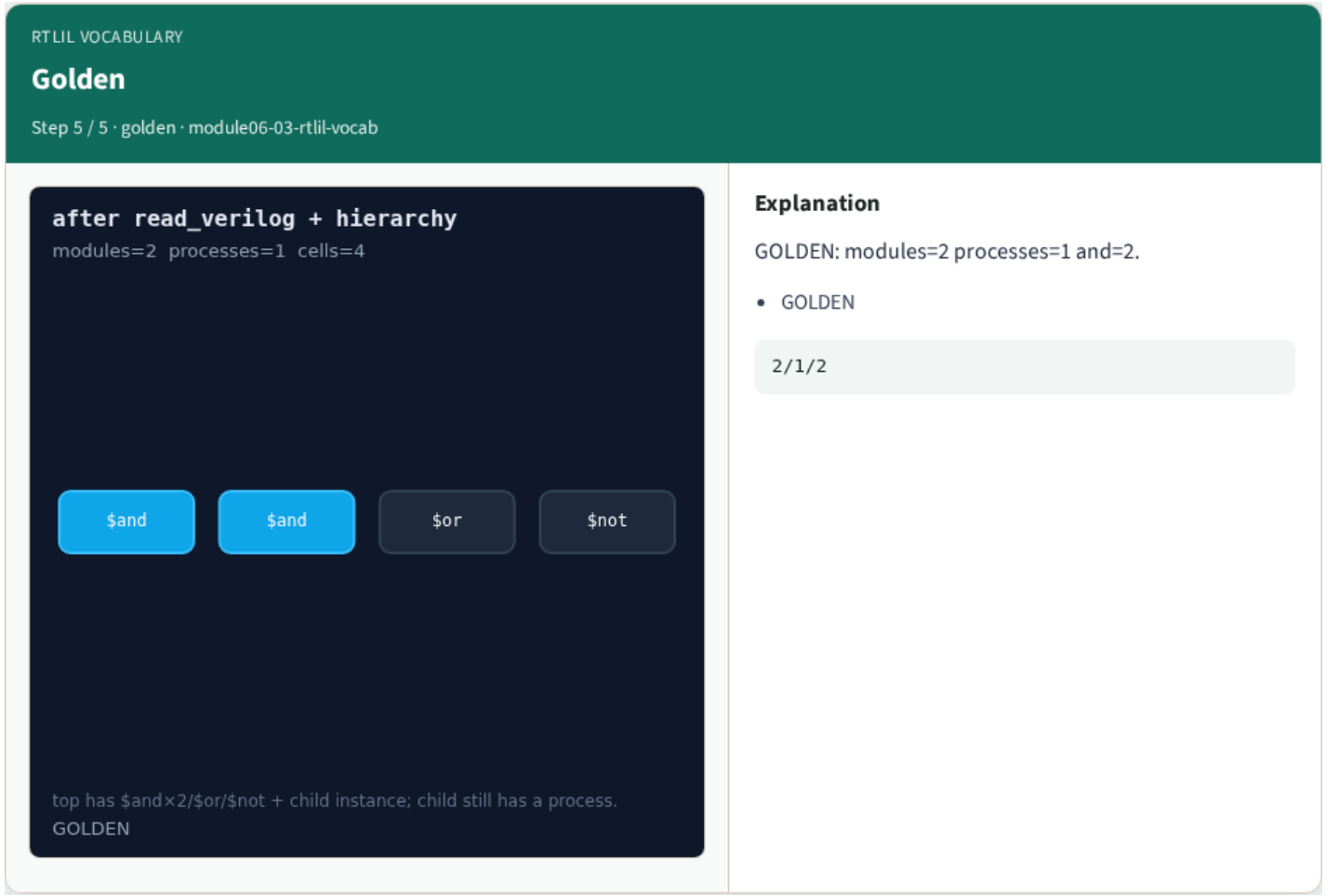
$and×2

$or and $not



$or and $not

Golden



Golden

Browser lab track

- Open **rtlil-vocab**
- Open **rtlil-vocab**
- Show read_hierarchy stage; select $and

Implement track

- Call pass_rtlil_vocab(); assert ok and and_count==2

Pitfalls

- Calling RTLIL 'gates' without $ prefix
- Ignoring processes

Try these

- From the common folder

Try these — try these

```
cd courses/learn_yosys/common
python3 real_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
load evidence after hierarchy
assert modules=2 processes=1 and=2
```

Next

- Complete the quiz for this part
- Next: real proc